

Closed-Loop PID Control of Light Intensity Using an LDR and WS2812 LED

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Abstract—This paper presents the design, implementation, and evaluation of a closed-loop PID control system for regulating light intensity using an LDR sensor and a WS2812 LED actuator. The system is implemented on an Arduino MKR1000 and incorporates advanced features including signal filtering, anti-windup mechanisms, rate limiting, and automatic calibration. The behaviour of both open-loop and closed-loop control systems is analysed and compared. Experimental results demonstrate that the closed-loop system provides significantly improved stability, accuracy, and disturbance rejection.

I. INTRODUCTION

Control systems are fundamental in engineering applications where maintaining a desired output in the presence of disturbances is required. Open-loop systems, while simple, lack the ability to adapt to environmental changes. In contrast, closed-loop systems utilise feedback to continuously correct their output, resulting in improved performance.

This project implements a closed-loop light control system using an LDR sensor to measure ambient light and a WS2812 LED to adjust illumination. A PID controller is used to regulate the system, and its performance is analysed in comparison to open-loop control.

II. APPLICATIONS AND SYSTEM CONTEXT

The implemented system has direct relevance to a range of real-world applications. In smart lighting systems, feedback-based control is used to maintain consistent illumination levels while minimising energy consumption. Similarly, mobile devices such as smartphones and laptops employ ambient light sensors to dynamically adjust screen brightness, improving both usability and battery efficiency.

From an engineering perspective, the system demonstrates how feedback control enables adaptability in dynamic environments. However, this adaptability comes at the cost of increased system complexity and the requirement for careful tuning and signal conditioning. The selection of an LDR and WS2812 LED provides a simple yet effective platform for demonstrating these principles, although the nonlinearity of the sensor and noise sensitivity introduce additional challenges that must be addressed.

III. SENSOR DATA ANALYSIS

The LDR operates as a resistive sensor whose resistance decreases nonlinearly as incident light intensity increases. When configured as part of a voltage divider, this varying resistance produces an analogue voltage that is sampled by the

Insert Sensor Raw vs Filtered Graph

Fig. 1. Comparison of raw and filtered LDR readings. The raw signal exhibits high-frequency noise and fluctuations, while the filtered signal demonstrates a smoother response with reduced variance. This figure illustrates the necessity of filtering in improving measurement reliability for control applications.

Arduino's 10-bit ADC, yielding values in the range 0–1023. However, due to the logarithmic response of the LDR, the relationship between light intensity and ADC output is not linear, which introduces additional complexity when designing a control system.

Raw sensor readings were observed to be highly variable, even under relatively constant lighting conditions. This variation arises from several sources, including electrical noise in the ADC, small fluctuations in ambient light, and quantisation effects inherent to digital conversion. These fluctuations are particularly problematic for control systems, as they can lead to unstable or oscillatory behaviour, especially when using derivative control.

To mitigate these issues, two stages of signal conditioning were implemented. Firstly, multiple ADC readings are averaged to reduce high-frequency random noise. Secondly, an exponential low-pass filter is applied to smooth the signal over time, as described by:

$$PV_f = PV_f + \alpha(PV - PV_f) \quad (1)$$

This filter acts as a first-order system, attenuating rapid fluctuations while preserving slower changes in light intensity. The choice of the parameter α determines the trade-off between responsiveness and smoothness. A higher value results in faster response but less filtering, whereas a lower value provides greater smoothing at the expense of delay.

The filtered signal is significantly more stable and suitable for feedback control. Without this filtering, the derivative term of the PID controller would amplify noise, leading to erratic actuator behaviour. Therefore, signal conditioning is essential for ensuring both stability and accuracy in the control system.

IV. ACTUATOR BEHAVIOUR

The WS2812 LED is a digitally addressable RGB LED that allows brightness control through an 8-bit value ranging from 0 to 255. The LED responds rapidly to input changes, making it suitable for real-time control applications. However, the perceived brightness of the LED is not linearly proportional

Insert Open-Loop Behaviour Graph

Fig. 2. Open-loop system response showing sensor readings over time while the actuator output remains constant. The graph illustrates how disturbances cause significant variation in measured light levels without any corrective action.

to the input value due to the nonlinear response of human vision, which can affect how changes in output are interpreted visually.

In this system, the actuator output is composed of a base output level combined with the PID controller output. The base output provides a steady illumination level around which the controller operates, ensuring that the system remains within an effective operating range. This is particularly important because very low brightness levels can result in poor sensitivity and increased noise in the sensor readings.

A key issue identified during implementation was the tendency for rapid fluctuations in LED brightness when the control signal changed abruptly. Although the PID controller may produce mathematically correct outputs, these rapid changes can result in visible flickering, which is undesirable in practical applications. To address this, rate limiting was introduced to constrain how quickly the actuator output can change between successive updates.

This rate limiting effectively smooths the actuator response, reducing high-frequency variations in brightness while maintaining overall control performance. As a result, the system achieves a balance between responsiveness and visual stability.

The behaviour of the actuator demonstrates that, in practical control systems, it is not sufficient to consider only the mathematical correctness of the control algorithm. Physical and perceptual constraints must also be taken into account to ensure acceptable real-world performance.

V. OPEN-LOOP CONTROL BEHAVIOUR

In the open-loop configuration, the LED is driven at a fixed brightness level that is independent of the sensor input. This means that the system does not incorporate any feedback and therefore has no mechanism to correct for changes in ambient light conditions.

Experimental observations show that when external disturbances are introduced, such as covering the sensor or introducing additional light sources, the measured sensor value changes significantly while the actuator output remains constant. This results in large deviations from the desired operating condition.

The graph would show the sensor value fluctuating over time in response to disturbances, while the LED output remains fixed. This highlights a fundamental limitation of open-loop control: the inability to adapt to environmental changes.

From a control engineering perspective, this demonstrates that open-loop systems are only suitable for environments where conditions are predictable and disturbances are minimal. In dynamic or uncertain environments, such as real-world lighting conditions, open-loop control is insufficient for maintaining consistent performance.

Insert Closed-Loop Step Response

Fig. 3. Closed-loop step response showing convergence to the setpoint. The graph illustrates how the system initially deviates from the target value and then settles with minimal steady-state error.

Insert Disturbance Response

Fig. 4. Disturbance response of the closed-loop system. The graph shows how the system reacts to a sudden change in ambient light and subsequently returns to the setpoint, demonstrating effective disturbance rejection.

VI. CLOSED-LOOP PID CONTROL

The closed-loop system employs a PID controller to regulate light intensity by continuously adjusting the actuator output based on the error between the desired setpoint and the measured sensor value. The control law is defined as:

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt} \quad (2)$$

The proportional term provides an immediate response to the error, ensuring that the system reacts quickly to deviations from the setpoint. The integral term accumulates past errors and eliminates steady-state offset, allowing the system to converge precisely to the desired value. The derivative term predicts future trends in the error, providing damping that reduces overshoot and improves stability.

In this implementation, the derivative is calculated based on the measured signal rather than the error. This approach reduces sensitivity to noise, as sudden changes in the error signal are often caused by measurement noise rather than true system dynamics. Additionally, derivative filtering is applied to further smooth the derivative term and prevent amplification of high-frequency noise.

A. Advanced Control Enhancements

Several enhancements were implemented to improve robustness. Anti-windup mechanisms prevent the integral term from growing excessively when the actuator reaches its limits. Without this, the system would exhibit large overshoots when it re-enters the controllable region. A deadband is also introduced to prevent unnecessary control action when the error is very small, reducing oscillations around the setpoint.

Automatic calibration allows the system to determine ambient conditions and adjust its setpoint and control direction accordingly. This ensures that the controller operates effectively across different lighting environments without requiring manual adjustment.

B. System Behaviour

The closed-loop system demonstrates stable convergence to the setpoint under normal operating conditions. When disturbances are introduced, the controller responds by adjusting the actuator output to counteract the change and restore the desired light level.

The step response graph would show the system gradually approaching the setpoint with limited overshoot, while the

disturbance graph would show a temporary deviation followed by recovery. These behaviours confirm that the controller is both stable and responsive.

Compared to open-loop control, the closed-loop system significantly reduces steady-state error and provides strong disturbance rejection. This demonstrates the effectiveness of feedback control in maintaining system performance under varying conditions.

VII. DESIGN METHODOLOGY AND EVALUATION

The system was developed using an iterative approach, beginning with a simple open-loop implementation to establish baseline behaviour. This was followed by the introduction of proportional control, which improved responsiveness but still resulted in steady-state error.

Integral control was then added to eliminate this error, although it introduced the risk of overshoot and instability. The derivative term was subsequently implemented to provide damping and improve system stability. During testing, it became clear that additional measures were required to address practical issues such as sensor noise and actuator limitations.

As a result, filtering, anti-windup mechanisms, and rate limiting were incorporated into the design. Automatic calibration was also introduced to improve adaptability. Each of these enhancements was validated experimentally, leading to a progressively more robust system.

This iterative development process highlights the importance of testing and refinement in control system design. Rather than relying solely on theoretical models, practical implementation revealed additional challenges that required engineering solutions.

VIII. USE OF SCIENTIFIC PRINCIPLES

The project applies fundamental principles of control theory, including feedback systems, PID control, and signal processing. These principles were used to solve a practical engineering problem involving real-world constraints such as noise and actuator limitations.

IX. INFORMATION SOURCES

The development of this system was informed by a combination of academic literature on control systems, technical documentation for hardware components, and experimental data obtained during testing. This combination of theoretical and practical sources enabled an effective and well-informed design process.

X. SOURCE CODE AND DEMONSTRATION

The full implementation is available at:

GitHub Repository:

<https://github.com/Thomriiii/Sensors-Actuators-Controllers/tree/main/Coursework1>

A demonstration of the system operation can be viewed at:

YouTube Video:

Insert YouTube Link Here

XI. CONCLUSION

A closed-loop PID control system has been successfully implemented and evaluated for regulating light intensity using an LDR sensor and WS2812 LED actuator. The system demonstrates clear advantages over open-loop control, including improved stability, reduced steady-state error, and effective disturbance rejection.

The inclusion of advanced features such as signal filtering, anti-windup, and rate limiting was essential in achieving reliable performance, highlighting the importance of addressing real-world constraints in control system design. The results demonstrate that feedback control provides a robust solution for maintaining consistent output in dynamic environments.

Overall, this project illustrates both the theoretical principles and practical considerations involved in implementing a PID controller, providing a comprehensive understanding of closed-loop control systems.